MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6814

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, subminiature, medium Mu

Outline	---	3-1 (EIA) (A)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	g	nc	h	nc	k	h	nc	p

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ik	(A) Ic	Pp	TE	(A) Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	°C	ft
Maximum: (see note 1)	6.6	275 (see note 2)	+5.5	200	---	1.0	22	5.5	2.2	+165	(see note 4)
Minimum:	6.0	---	-55	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	0	150	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS			UNIT
						MIN	BOGEY	MAX	
	Quality conformance inspection, part 1								
(A) 1256	Electrode current (1) (anode)	Eb = 125 Vdc; Rk = 0; Ec = 3 Vdc; Rp = 2,700 ohms; Rg = 0.06 Meg (see note 5)	0.4	II	Ib	14	15.5	---	mAdc
(A) 1261	Electrode voltage (grid)	Eb = 140 Vdc; Rp = 2,700 ohms; Rg = 0.06 Meg; Ec/Ib = 100 μAdc	0.4	II	Ec	---	---	-8.5	Vdc
(A) 1266	Total grid current	Eb = 150 Vdc; Rk = 220 ohms; Rg = 1.0 Meg (see note 5)	0.4	II	Ic1	0	---	-0.4	μAdc
(A) 1301	Heater current		0.4	II	If	138	---	162	mA
(A) 1336	Heater-cathode leakage		0.4	II	Ihk	---	---	5.0	μAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---	---
(A) Appendix D, 30(a), 40, 50	Visual and mechanical inspection criteria		---	---	---	---	---	---	---
(A) Appendix D, 30(b)	Dimensions	Outline 3-1; dimensions A, D, and E	1.0	I	---	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		2.5	I	R	50	---	---	Meg
1031	Low-frequency vibration	F = 40 Hz; 15 G; Rp = 10,000 ohms; Ck = 1,000 μF	2.5	I	Ep	---	---	50	mVac
(A) 1256	Electrode current (2) (anode)	Ef = 5.7 V; Eb = 125 Vdc; Ec = 3 Vdc; Rp = 2,700 ohms; Rg = 0.06 Meg; Rk = 0	2.5	I	Ib	13	---	---	mAdc
1256	Electrode current (3) (anode)		6.5	S3	Ib	7.5	---	12.5	mAdc
1266	Grid emission	Ef = 7.5 V; Rg1 = 1.0 Meg; Ec1 = -8.5 Vdc (see note 6)	2.5	I	Ic1	0	---	-0.5	μAdc
1306	Transconductance		2.5	I	Sm	5,000	---	7,000	μmhos
1316	Amplification factor		6.5	S3	Mu	25	---	33	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1331	Direct-interelectrode capacitance	No shield	6.5	Code F	Cgp	1.1	---	1.8	pF
					Cin	1.6	---	2.8	pF
					Cout	0.5	---	0.9	pF
1116	Lead fatigue		2.5	Code F	---	---	---	---	---
(A) 2126	Glass strain		4.0	I	---	---	---	---	---
1041	Shock	450 G (see note 8)	---	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 7	---	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	---	Ep	---	---	200	mVac
		Heater-cathode leakage	---	---	Ihk	---	---	15	μAdc
		Electrode current (1) (anode)	---	---	Ib(1)	13	---	---	mAdc
		Electrode voltage (grid)	---	---	Ec	---	---	-8.5	Vdc
(A) 1105	Permanence of marking		---	---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1 ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; Eb = Ec = 0 (see note 9)	2.5	Code H	---	---	---	---	---	---
1516	Stability life	Ef = 6.3 V; Eb = 250 Vdc; Ec = 0; Ehk = 200 Vdc; Rk = 1,000 ohms; Rg = 1.0 Meg; TA = room	1.0	Code I	---	---	---	---	---	---
---	Life-test end point (stability) (2 and 20 hours)	Change in anode current (1) of individual tubes	---	---	---	---	ΔIb t	---	10	%
1521	Survival-rate life	Stability life test, or equivalent conditions	---	II	---	---	---	---	---	---
---	Life-test end points (survival-rate) (100 hours)	Short and discontinuity detection	0.65	---	---	---	---	---	---	---
		Electrode current (1) (anode)	1.0	---	---	---	Ib	13	---	mAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
1501	Life-test provisions (intermittent)	Survival rate life-test conditions TE = +165° C (min) (see note 10)	---	---	---	---	---	---	---	---
---	Life-test end points (intermittent) (500 hours)	Inoperatives	---	---	1	3	---	---	---	---
		Electrode current (1) (anode)	---	---	1	3	Ib	13	---	mAdc
		Electrode voltage (grid)	---	---	1	3	Ec	---	-8.5	Vdc
		Electrode current (grid)	---	---	1	3	Ic	0	-0.6	μAdc
		Heater-cathode leakage	---	---	2	5	Ihk	---	10	μAdc
		Heater current	---	---	2	5	If	136	166	mA
		Insulation of electrodes	---	---	2	5	R	25	---	Meg
		Total defectives	---	---	4	8	---	---	---	---
---	Life-test end points (intermittent) (1, 000 hours)	Inoperatives	---	---	2	5	---	---	---	---
		Electrode current (1) (anode)	---	---	2	5	Ib	13	---	mAdc
		Electrode voltage (grid)	---	---	2	5	Ec	---	-8.5	Vdc
		Electrode current (grid)	---	---	2	5	Ic	0	-0.6	μAdc
		Insulation of electrodes	---	---	3	6	R	25	---	Meg
		Total defectives	---	---	5	10	---	---	---	---
1511	Cathode interface life (500 hours)	Ef = 6.9 V; other electrodes disconnected (see note 11)	---	---	---	---	---	---	---	---
---	Life-test end points (cathode interface) (500 hours)	Ef = 5.7 ±0.05 Vdc; Eb = 100 Vdc; Ec/Ib = 1.0 mAdc (see note 3)	---	---	1	3	Ri	---	50	Ohms

NOTES:

1. The absolute-maximum ratings for pulse characteristics apply to a square-topped pulse of 1, 000 Hz frequency and 1.0 percent duty factor.
2. The requirements of "Anode voltage" shall apply with the following exception: When the load impedance is of such type that the instantaneous voltage at the anode never exceeds the supply voltage, the supply voltage may be 300 Vdc, provided the maximum rated average dissipation is never exceeded on any electrode.
3. Preheat approximately 5 minutes prior to testing, using Ef = 6.0 V; other electrodes disconnected. No other test shall be made from the start of the cathode interface life test until after the measurement of the end-point characteristic, following completion of the indicated minimum number of life-test hours.
4. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
5. This test to be performed at the conclusion of the holding period.
6. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test immediately after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Eb Vdc	Ec Vdc	Rk Ohms	Rg1 Meg
7.5	150	0	220	1.0

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NOTES: -Continued

(A) 7. This test shall be conducted on the initial lot and therefore on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

8. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.

9. The no-load to steady-state full-load regulation of the heater voltage supply shall be not more than 3.0 percent. This test shall be made on a lot-by-lot basis.

(A) 10. Envelope temperature (TE) requirements when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

11. The life-test sample shall consist of 20 tubes, and not more than 2 tube failures shall be permitted. In the event of rejection of the first sample due to failure of more than 2 tubes, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed five. A life-test defect is defined as a failure to meet the life-test end-point limits as specified on the tube specification sheet. The life-test sample shall be read at zero hours and 500 hours (plus 48 hours, minus 24 hours).

The margins of this tube specification sheet are marked with a revision letter (A) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Air Force - 11

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